

## **CPU REGISTERS OVERVIEW**

### **1. General Purpose Registers (GPRs)**

- Purpose: Hold operands, intermediate results, or temporary data during program execution.
- Examples (x86 16-bit): AX, BX, CX, DX
- Sub-parts: Can be accessed as high/low bytes (e.g., AX → AH, AL)
- Usage: Arithmetic operations, data transfer, loop counters (CX commonly used)
- Analogy: Like scratch paper for doing calculations.

### **2. Segment Registers**

- Purpose: Hold addresses of memory segments.
- Examples: CS, DS, SS, ES
- Usage: Allow CPU to access different memory areas.
- Analogy: Bookmarks pointing to different sections of a book.

### **3. Pointer & Index Registers**

- Purpose: Hold memory addresses for data and instruction access.
- Examples: SP, BP, SI, DI
- Usage: Stack operations and string/data movement.
- Analogy: A finger pointing to where data lives.

### **4. Special Purpose Registers**

- Purpose: Control and status management.
- Examples: IP, FLAGS Register
- Usage: Manages program flow and indicates operation results.
- Analogy: IP = roadmap; FLAGS = traffic signals.

### **5. Floating Point Registers**

- Purpose: Floating point arithmetic.
- Examples: ST0–ST7
- Usage: Scientific and complex math operations.
- Analogy: A calculator dedicated to decimals.

### **6. Control Registers**

- Purpose: Configure and manage CPU operations.
- Examples: CR0–CR4
- Usage: Used by OS for memory protection and mode switching.

- Analogy: CPU settings panel.

## **7. Instruction & Status Registers**

- Purpose: Assist in instruction processing.
- Examples: IR, MAR, MBR
- Usage: Temporarily hold instructions and data.
- Analogy: Conveyor belt in a factory.

## **8. Miscellaneous / Temporary Registers**

- Purpose: Used internally for calculations.
- Examples: Temporary ALU registers.
- Analogy: Extra hands to hold items while working.

### **Summary:**

Registers act as ultra-fast temporary storage within the CPU to speed up instruction execution.